MONICA R

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ABOUT

Enthusiastic Software Developer with a strong foundation in programming and problem-solving. Skilled in Java, , Python and basic software development concepts. Eager to apply technical knowledge to build efficient applications and grow in a challenging development environment.

Education

IFET College Of Engineering Villupuram, BE in Electronics and Communication Engineering Nov 2022 – May 2026

- CGPA: 8.17/10

Amalorpavam Higher Secondary School,Pondicherry. Jun 2021 – May 2022

- HSC Percentage: 79.5

Experience

INTERN – Web Developer – *LeMenz Technologies Pvt. Ltd., Pondicherry* 2024

- Gained hands-on experience in frontend and backend development through real-time projects.
- Improved skills in UI design, API integration, and web/application development frameworks.

INTERN – Python Full Stack Developer – *VEI Technologies Pvt. Ltd., Tindivanam* 2025

- Strengthened backend development skills using Python and frameworks such as Django and Flask.
- Acquired practical experience in database management, SQL queries, and frontend–backend integration.

INPLANT TRAINING - Embedded Systems & IoT Trainee – *TVS Training Services Ltd., Chennai* 2025

- Worked with microcontrollers, sensors, and electronic circuit design in real-world scenarios.
- Developed skills in Embedded C, firmware testing, and hardware debugging.
- Learned PCB fundamentals, wiring techniques, and embedded system problem-solving. .

INTERN – Graduate Engineer – *Eaton Power Quality Pvt Ltd ,Setharapet, Pondicherry* 2026

- Assisted in testing and monitoring power quality systems for industrial applications.
- Supported troubleshooting and maintenance activities of electrical and power electronic equipment.
- Collaborated with engineers in analyzing system performance and preparing technical reports.
- Gained practical exposure to UPS systems, power distribution, and electrical safety practices.

Publications

1. Arduino-Based Alcohol Detection Device: Enhancing Safety in Vehicle Operation through Sensor Technology, March 2025

Asian Journal of Applied Science and Technology, ***Vol. 8, Issue 1***

2. A Novel and Reliable SOS Alert Band System for Women Safety, March 2024

Asian Journal of Applied Science and Technology, ***Vol. 9, Issue 1***

Projects

ARDUINO BASED TOLL TAX SYSTEM 2023

- Developed an automated toll collection system using Arduino with sensors (IR/RFID) for vehicle detection and identification.
- Implemented real-time toll calculation and automatic transaction handling to ensure fast and accurate processing

- Integrated LCD display and servo motor barrier mechanism to streamline traffic flow and enhance operational efficiency
- Designed an embedded system architecture combining **sensor integration**, **microcontroller programming**, and **automation control** for smart transportation applications.
- Tools Used: Arduino Idle, Microcontroller

VIRO SHIELD - INTELLIGENT DOOR AUTOMATION SYSTEM 2024

- Initiated and led the creation of an automated entry control system integrating temperature screening sensors to enhance security and health safety.
- Managed hardware and software development efforts ensuring project success
- Enhanced safety and energy efficiency by integrating alert signals and auto-close functionality based on sensor feedback.
- Tools Used: IOT, Arduino Idle

SECURESHE - WERABLE SAFETY DEVICE 2025

- Conceptualized and Led the design of an offline GSM-based wearable safety device with nerve sensors on the foot thumb for discreet panic detection. The device triggers automatic SOS alerts using biometric and environmental sensors without mobile dependency.
- Enhanced user safety with quick-response mechanisms, LED/Vibration indicators, and reliable real-time activation.

CAPACITY ANALYSIS OF MIMO-OFDM WIRELESS COMMUNICATION SYSTEM UNDER RAYLEIGH FADING CHANNEL USING ADAPTIVE SNR OPTIMIZATION 2026

- Developed a **MIMO-OFDM Wireless Communication System** under **Rayleigh Fading Channel** for reliable and high-speed wireless data transmission.
- Designed and simulated a **Microstrip Patch Antenna** using Keysight ADS to improve **signal gain**, **bandwidth**, and communication efficiency.
- Implemented **Adaptive SNR Optimization** to enhance **channel capacity**, **spectral efficiency**, and overall wireless network performance.
- Developed an interactive dashboard using Streamlit for real-time visualization and monitoring of wireless communication performance metrics, supporting next-generation **5G communication systems**.
- Enhanced user safety with quick-response mechanisms, LED/Vibration indicators, and reliable real-time activation.
- Tools used: ADS software,VS code, Streamlit run app.py

Technologies

Languages: Java, Python, C , C++,SQL, JavaScript , HTML, CSS

Technologies: .NET, Node.js

Certifications

1. ***Java Full Stack Certification – Wipro***
2. ***Drone Technology Certification – Atal Incubation Center, Pondicherry***
3. **CCNA – Cisco Certified Network Associate**